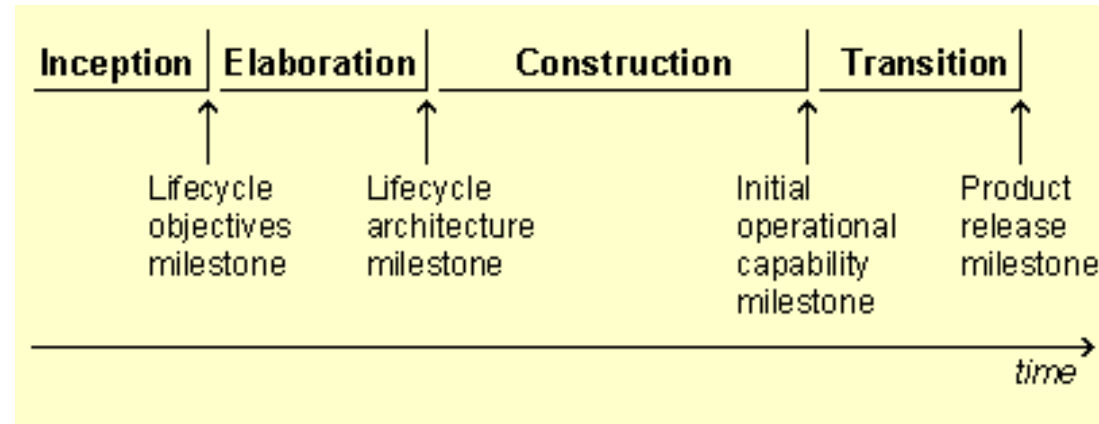


Project elaboration phase

Projeto em Engenharia de Computadores e Informática – (LECI), Departamento de Eletrónica, Telecomunicações e Informática, Universidade de Aveiro

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Phases and milestones of a project (OpenUP)



source:

http://epf.eclipse.org/wikis/openup/practice.mgmt.risk_value_lifecycle.base/guidances/concepts/phase_milestones_5678231E.html

elaboration phase

The purpose is to establish the baseline of the architecture of the system and provide a stable basis for the bulk of the development effort in the next phase.

Get a more detailed understanding of the requirements.

*Design, implement, validate, and establish the baseline for the architecture. **Design, implement, and test a skeleton structure of the system.***

Mitigate essential risks and produce accurate schedule.

source: http://epf.eclipse.org/wikis/openup/practice.mgmt.risk_value_lifecycle.base/guidances/concepts/elaboration_phase_BE880435.html

Simple and easy to follow recommendations that largely pay-off

Organize documents and resources into shared folders

Take detailed notes of everything and record them in a notebook even in raw form

Avoid big designs with no [code] evidence

Elaboration phase: requirements elicitation

Describe the requirements of system users, customers and other stakeholders

1. Requirements gathering

Sources of information including interviews, questionnaires, user observation, workshops, brainstorming, etc.

2. Functional requirements

What the system is supposed to accomplish, for example: business Rules, administrative functions, authentication and authorization levels, reporting, historical data, or Regulatory Requirements.

3. Context and State of The Art (SOA)

How the system is expected to be used and what has been done in this context

Elaboration phase: requirements elicitation

4. Actors / Personas

Describe the target users in your system.

5. Use cases / User stories

Describe the interaction between users and the system to achieve specific goals. Tip: organize the use cases by packages.

<http://agilemodeling.com/artifacts/useCaseDiagram.htm>

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-use-case-diagram/>

6. Non-functional requirements

Describe how the system works (while functional requirements describe what the system should do), for example: performance (e.g., response time, capacity , availability , recoverability, maintainability , security , data integrity , usability, or interoperability).

do not forget assumptions and dependencies.

Elaboration phase: system architecture

Deployment diagram

Describe the runtime processing modes and the components that live on them.

Include the technologies used in the system.

<http://www.agilemodeling.com/artifacts/deploymentDiagram.htm>

<https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-deployment-diagram/>

Architecture baseline (executable)

Repository with initial structure; a minimal end-to-end slice running or a proof of concept.

[Domain model]

A diagram capturing the main entities and relationships between them may also help in some systems.

<http://agiledata.org/essays/agileDataModeling.html#InitialDomainModel>

[User interaction]

Design mock-ups as soon as possible.

[...] may not apply to all projects

recommended to do list → milestone 2

Present a detailed State-of-the-Art and how your project relates to previous work

Requirement's elicitation

Define goals

Collect information: documents, interviews, etc

Functional and non-functional requirements

Actors, uses cases and other

System Architecture

*Skeleton structure of the system **is tested**: system components must communicate with each other*

Design mock-ups as soon as possible

Milestone 2

1. Prepare a **12 minutes** presentation:

Details on elaboration phase results are mandatory

A demonstration to show that the main components of the architecture work together, a proof of concept, or equivalent, is required.

2. Prepare the first part of the technical report: a **maximum of 15 pages** is recommended. An example of the report structure:

Abstract

Introduction

Related work, tools and technologies

Requirements elicitation and architecture

Expected results

See also Technical Report example, chapters 1,2 and 3, in e-Learning.